

6. Naive Bayes Algorithm

Aim

To implement the Naive Bayes classification algorithm using Python and evaluate its performance using accuracy and a confusion matrix.

Software Used:

Google Colab

Procedure

1. Open Google Colab and create a new notebook.
2. Import the required Python libraries.
3. Load the Iris dataset.
4. Divide the dataset into training and testing data.
5. Create a Gaussian Naïve Bayes classifier.
6. Train the classifier using the training data.
7. Predict the classes of the testing data.
8. Calculate the accuracy.
9. Generate the confusion matrix.
10. Display the confusion matrix and actual vs predicted graph.

Code

```
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score

from sklearn.metrics import confusion_matrix

from sklearn.metrics import ConfusionMatrixDisplay

# Load dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create Naive Bayes model

model = GaussianNB()

# Train model

model.fit(X_train, y_train)

# Prediction
```

```
y_pred = model.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

print("Naive Bayes Classification")

print("Accuracy:", accuracy)

# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)

# Display confusion matrix

disp = ConfusionMatrixDisplay(

    confusion_matrix=cm,

    display_labels=iris.target_names

)

disp.plot()

plt.title("Naive Bayes Confusion Matrix")

plt.show()

# Actual vs Predicted graph

plt.figure(figsize=(7,5))

plt.scatter(range(len(y_test)), y_test,

            label="Actual", marker="o")
```

```
plt.scatter(range(len(y_pred)), y_pred,  
            label="Predicted", marker="x")  
  
plt.xlabel("Test Sample")  
  
plt.ylabel("Class")  
  
plt.title("Naive Bayes Actual vs Predicted")  
  
plt.legend()  
  
plt.grid()  
  
plt.show()
```

OUTPUT

Naive Bayes Classification

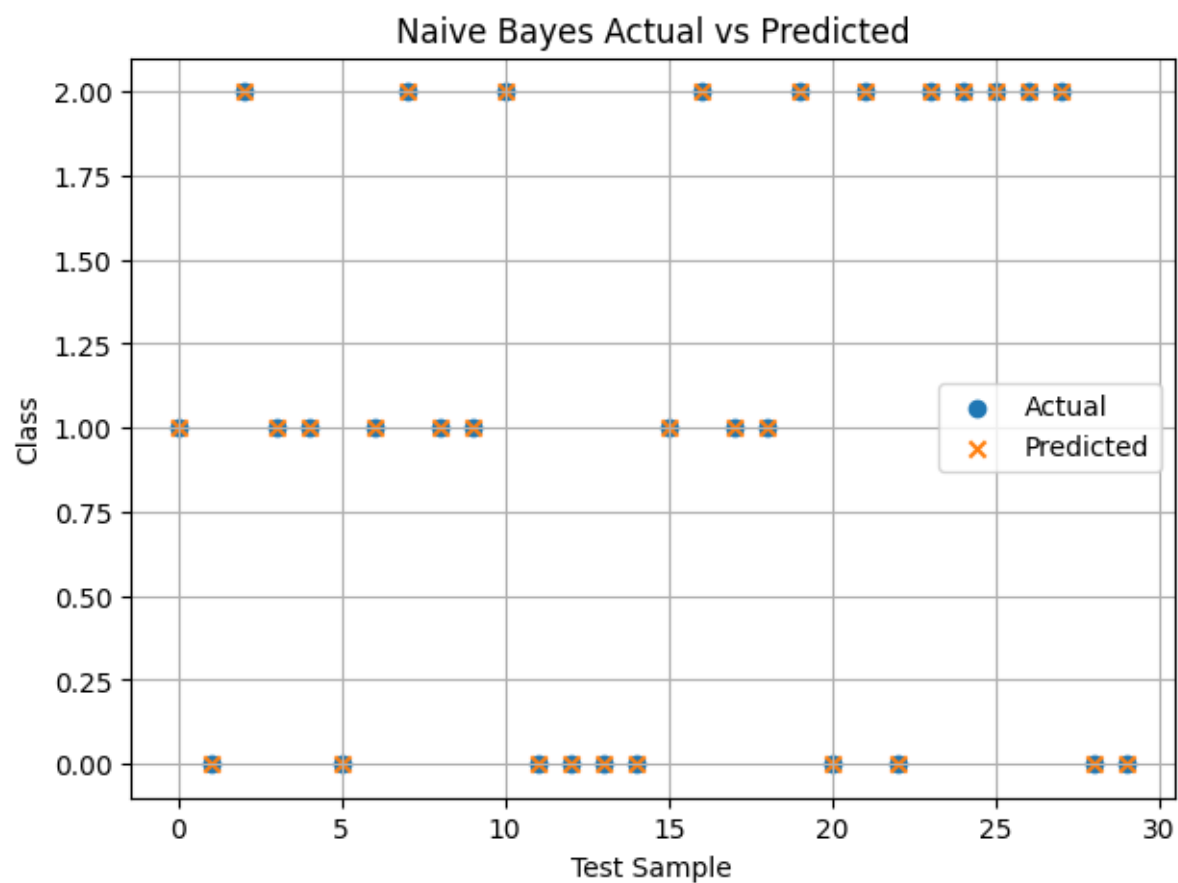
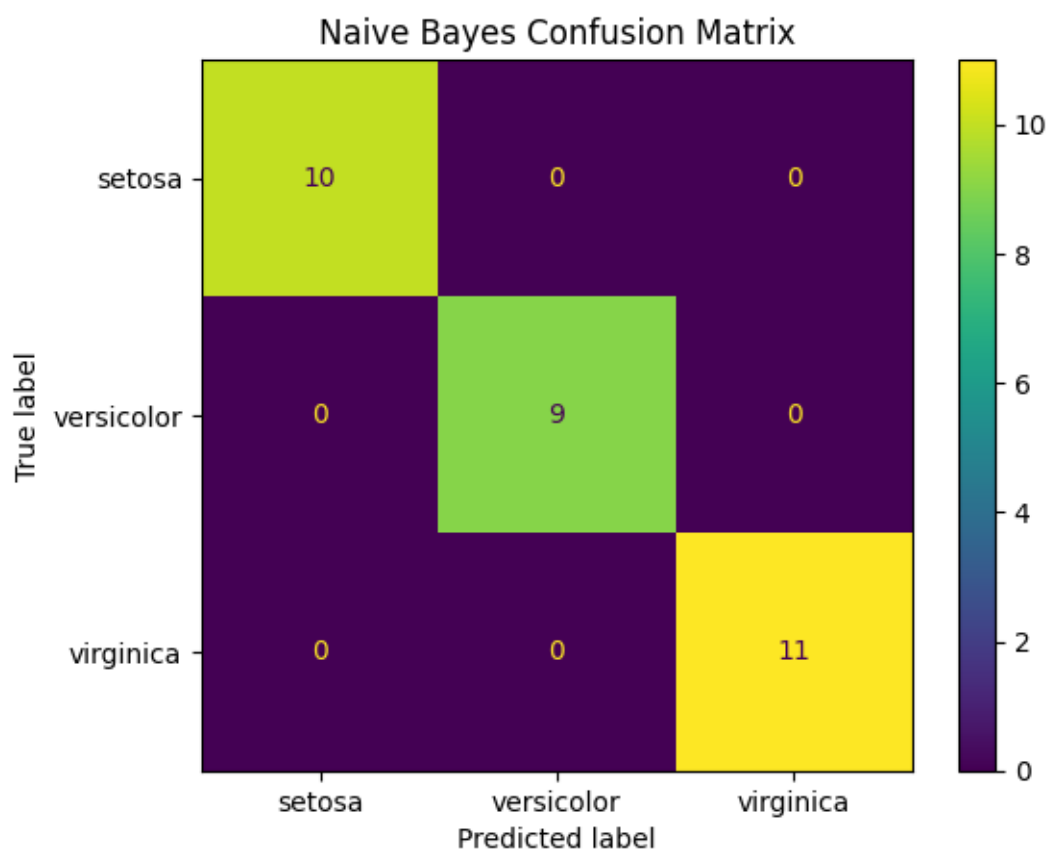
Accuracy: 1.0

Confusion Matrix:

```
[[10  0  0]
```

```
 [ 0  9  0]
```

```
 [ 0  0 11]]
```



Result

Thus, the Naïve Bayes classification algorithm was successfully implemented using Python in Google Colab, and the performance was evaluated using accuracy and confusion matrix.